

Chat filter accuracy: unanchored substring matching

Ragnarok: Rebirth Global · independent measurement against the live service · 12 September 2026

The filter matches its blocked terms as **unanchored substrings**, and the term list is predominantly **Brazilian Portuguese**. Short Portuguese entries applied to all eight locales censor ordinary vocabulary in every language the game ships in.

THE SHORTEST DEMONSTRATION

thank **delivered** thanks **refused**
one letter apart; the plural creates the sequence ks

THE RULE IS TWO CHARACTERS, MATCHED ANYWHERE

cu	refused	xxcuxx	refused	xxcxx	ok	xxuxx	ok	xxucxx	ok
ks	refused	xxksxx	refused	xxkxx	ok	xxsxx	ok	xxskxx	ok

Refused bare and inside inert padding, at any position. Neither letter is refused alone; neither reversed pair is refused. Across **10,627 accepted words**, none contains **cu** or **ks**.

CONFIRMED RULES AND ORDINARY WORDS AFFECTED

RULE	WORDS	ORDINARY WORDS CENSORED
cu	5,016	document, discuss, security, focus, circus
nb	956	number, inbox, unbind
meter	695	parameter, diameter, kilometer
anta	419	advantage, santa, fantastic
ks	408	thanks, tasks, books, works, weeks
pus	230	campus, octopus, push
pica	227	typical, tropical
puta	95	reputation, computation

8,414 distinct ordinary English words, 3.6% of the language

DELIVERED UNDER THE SAME TEST

bitch	ok	rape	ok		
sex	ok	slut	ok	whore	ok

All 27 Portuguese terms tested are refused, including *arrombado* and *otario*. The filter is not over-aggressive in general — it is aggressive about one language's vocabulary and largely absent for another's. One root cause: a term list never reviewed against the locales it is applied to.

THREE WAYS TO FIX IT, IN INCREASING EFFORT

- 1 Allow the confirmed words.** `findings/words-to-allow.txt` lists **342 ordinary words** confirmed refused, each annotated with the rule responsible. If the filter already supports an exception list, this is a data change and nothing else — it removes false positives only.
- 2 Require a word boundary for short terms.** One rule — *entries shorter than five characters match only as whole words* — removes most of the damage. `cu` still refuses `cu` and stops refusing `document`. Entries of five or more characters are untouched, so `caralho` matches exactly as it does today.
- 3 Replace the matcher.** `appendix/reference-engine/` — per-term match modes, severity tiers, allowlist rescue, evasion folding. No dependencies, MIT, 53 tests and 29 conformance vectors, at the same `check(text)` seam as today's filter.

15,307 messages sent through the ordinary game client into a private party channel; no client modification, no server access. **342 ordinary words confirmed refused**, **10,627 confirmed delivered**. Every rule was isolated in inert padding before being reported and every finding required two independent observations; the measured per-probe error rate is 1.2%. Logs: `findings/raw-logs/`.